

Linear regression I: normal equation and geometry

The model $y = X\theta + \epsilon$, least squares, the closed-form solution, two worked examples and the projection picture.

THE QUESTION

Given N examples, which θ makes $\hat{y} = X\theta$ closest to y , and can we write that θ down directly?

THE INTUITION

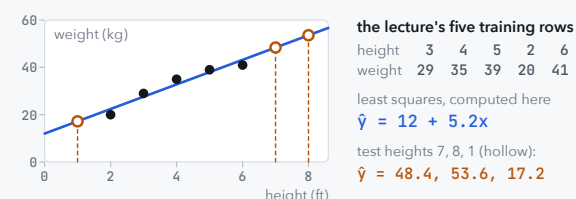
Score a fit by the sum of squared residuals $\epsilon^T \epsilon$. It is a bowl in θ , and setting its gradient to zero gives linear equations. Geometrically, $X\theta$ can only reach the column space of X , and the best $\hat{y}$ is the point there nearest to y .

KEY TAKEAWAY

$\hat{\theta} = (X^T X)^{-1} X^T y$ when the columns of X are independent. The residual is at right angles to every column: $X^T(y - X\hat{\theta}) = 0$.

data $(x_i, y_i) \rightarrow$ **X with a column of ones** $\rightarrow$ **minimise $\epsilon^T \epsilon$** $\rightarrow$ **$X^T X \theta = X^T y$** $\rightarrow$ **solve for θ , predict $\hat{y} = X\theta$**

1 A line has an intercept and a slope



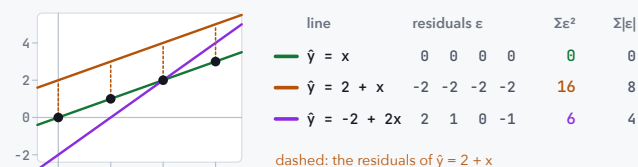
The lecture's task: predict weight from height, with five training rows and three test heights. The model is $\text{weight}_i \approx \theta_0 + \theta_1 \cdot \text{height}_i$, the same shape as $v = u + at$. The output is continuous, so this is regression. The blue line is the least-squares fit this sheet derives.

2 Stack N equations into $\hat{y} = X\theta$

$\hat{y} \ 5 \times 1$	$X \ 5 \times 2$	$\theta \ 2 \times 1$	row i: $\hat{y}_i = \theta_0 \cdot 1 + \theta_1 \cdot \text{height}_i$ θ_0 intercept (bias), θ_1 slope
$\hat{y}_1$	1 3	θ_0	M features: X is $N \times (M+1)$ $\hat{y} \ (N \times 1) = X \ (N \times (M+1)) \cdot \theta \ ((M+1) \times 1)$ water demand, two features: row i = [1, temp_i, occupants_i]
$\hat{y}_2$	1 4	θ_1	
$\hat{y}_3$	1 5		
$\hat{y}_4$	1 2		
$\hat{y}_5$	1 6		

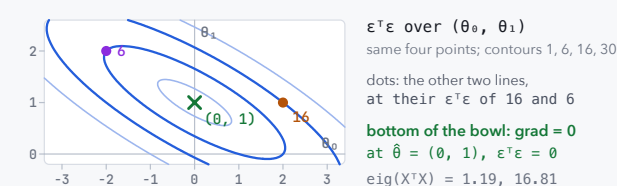
Each row of X is one sample and starts with a 1, which multiplies the intercept θ_0 . With M features, $\hat{y}_i = \theta_0 + \theta_1 x_{i,1} + \dots + \theta_M x_{i,M} = x_i^T \theta$ with $x_i^T = (1, x_i)$, a column, hence the transpose. For water demand from temperature and occupants we expect $\theta_1, \theta_2 > 0$ and a positive base demand θ_0 .

3 Least squares adds up squared residuals



The model is $y = X\theta + \epsilon$, so $\epsilon_i = y_i - (\theta_0 + \theta_1 x_i)$. A good fit keeps every $|\epsilon_i|$ small. Least squares minimises $\epsilon_1^2 + \dots + \epsilon_N^2$ (the squared L2 norm); the L1 alternative minimises $\sum |\epsilon_i|$. The lecture assumes ϵ_i i.i.d. $\mathcal{N}(0, \sigma^2)$. Its checklist before using the model: a linear relationship, independent observations, constant error variance (homoscedasticity), normal errors (for inference) and no multicollinearity.

4 A zero gradient gives the normal equation



Expand the objective; the two cross terms are equal scalars:

$$\epsilon^T \epsilon = (y - X\theta)^T (y - X\theta) = y^T y - 2y^T X\theta + \theta^T X^T X\theta.$$

With the maths sheet's rules, $\nabla a^T \theta = a$ and $\nabla \theta^T A \theta = 2A\theta$ for symmetric $A = X^T X$:

$$\nabla_{\theta} \epsilon^T \epsilon = -2X^T y + 2X^T X\theta = 0 \Rightarrow \hat{\theta} = (X^T X)^{-1} X^T y.$$

The inverse exists when the columns of X are independent (panel 11). Then $X^T X$ has positive eigenvalues (1.19 and 16.81 in panel 5's example), the bowl has one bottom, and the zero-gradient point is the minimum.

5 Worked example: four points on $y = x$

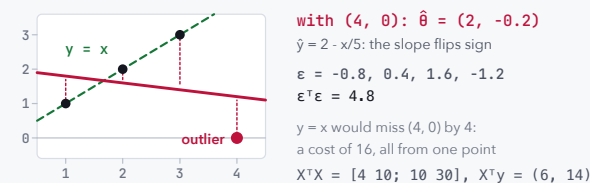
X	y	$X^T X$	$X^T y$	$(X^T X)^{-1}$	$\hat{\theta}$
1 0 0	0	4 6	6	1/20	14 -6 0
1 1 1	1	6 14	14		-6 4 1
1 2 2	2				
1 3 3	3				

det $X^T X = 4 \cdot 14 - 6 \cdot 6 = 20$
 $\hat{\theta} = (1/20)(14 \cdot 6 - 6 \cdot 14, -6 \cdot 6 + 4 \cdot 14) = (0, 1)$

$\hat{y} = x$
all four points on it

For $x = (0, 1, 2, 3)$ and $y = x$, $X^T X$ has determinant 20 and $\hat{\theta} = (0, 1)$: the fit is $\hat{y} = x$ with zero error, as on the slides. On page 2, y here lies in the column space of X .

6 One outlier tilts the whole line

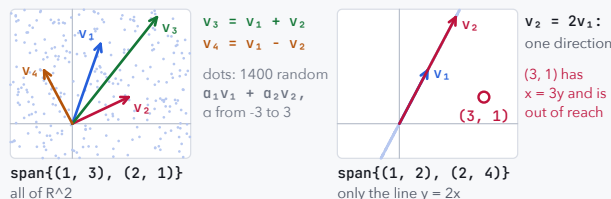


The first three points alone give $\hat{\theta} = (0, 1)$. Adding $(4, 0)$ gives $\hat{\theta} = (2, -1/5)$, as on the slides. Squaring makes one large residual cost more than several small ones, so least squares pulls the line towards the outlier. Plot the data before trusting a fit.

Least squares projects y onto the column space of X

$X\theta$ can only reach combinations of the columns of X . The best $\hat{y}$ is the nearest such point, and the residual is perpendicular to all of them.

7 A span is every linear combination



A linear combination of $v_1, \dots, v_k$ is $\alpha_1 v_1 + \dots + \alpha_k v_k$ with real α_j ; the span is the set of all of them. With the vectors as the columns of V , the span is $\{V\alpha\}$. Two independent vectors in $\mathbb{R}^2$ span the plane; parallel ones span a line.

8 $X\theta$ lives in the column space of X

X	y	
1	2	8.90
1	-2	0.61
1	2	1.78

$\bar{x}_1 \quad \bar{x}_2$

$$X\theta = \theta_0 \bar{x}_1 + \theta_1 \bar{x}_2 = (\theta_0 + 2\theta_1, \theta_0 - 2\theta_1, \theta_0 + 2\theta_1)$$

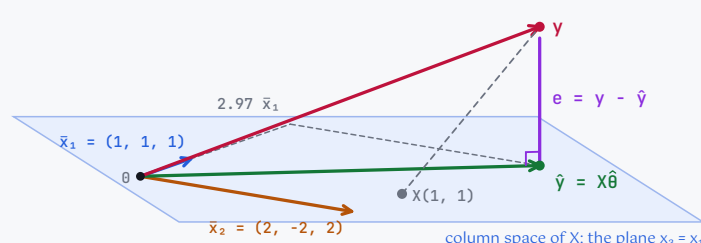
entry 1 equals entry 3 for every θ :
the column space is the plane $x_3 = x_1$

y has $y_1 = 8.90$ but $y_3 = 1.78$: off the plane,
so no θ makes $X\theta = y$ exactly

$X\theta = \theta_0 \bar{x}_1 + \theta_1 \bar{x}_2 + \dots$ combines the columns $\bar{x}_j$ of X , so every prediction vector $\hat{y}$ lies in their span, a subspace of $\mathbb{R}^N$. Least squares picks

$$\hat{y} = \underset{\hat{y} \in \text{span}\{\bar{x}_j\}}{\operatorname{argmin}} \|y - \hat{y}\|_2.$$

9 $\hat{y}$ is the projection of y : the residual is perpendicular to every column



$$\hat{\theta} = (X^T X)^{-1} X^T y = (2.974, 1.181)$$

$$\hat{y} = X\hat{\theta} = (5.336, 0.613, 5.336)$$

$$e = y - \hat{y} = (3.560, 0, -3.560)$$

$$X^T e = (0, 0) \text{ up to } 9e-16: e \text{ at } 90^\circ \text{ to both columns}$$

$$\|y - \hat{y}\|^2 = 25.34: \text{no point of the plane is closer to } y$$

$$\text{another point of the plane, } X(1, 1) = (3, -1, 3):$$

$$\|y - X(1, 1)\|^2 = 25.34 + 13.51 = 38.86 \text{ (Pythagoras)}$$

The nearest point of a plane to y is the foot of the perpendicular dropped from y . So $e = y - \hat{y}$ is at right angles to each column: $\bar{x}_j^T (y - \hat{y}) = 0$ for every j , which in matrix form is $X^T (y - X\theta) = 0$, the normal equation $X^T X\theta = X^T y$ again, reached without calculus. So $\hat{y} = X(X^T X)^{-1} X^T y$: that matrix projects any y onto the column space. The numbers are the lecture's three-point example (geometric notebook); the sketch is drawn in axes along the plane and along its normal.

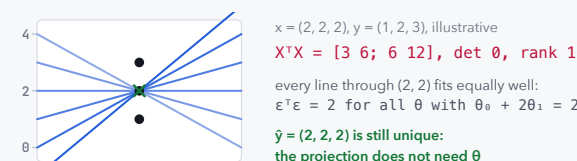
10 $X^T e = 0$ gives quick checks

outlier fit: e and the columns of X

e	-0.8	0.4	1.6	-1.2	every column meets e at 90°
$\bar{x}_1$	1	1	1	1	$\bar{x}_1^T e = 0$: residuals sum to 0
$\bar{x}_2$	1	2	3	4	$\bar{x}_2^T e = -0.8 + 0.8 + 4.8 - 4.8 = 0$

The column of ones makes row 1 of $X^T e = 0$ read $\sum_i e_i = 0$: with an intercept the residuals sum to zero, and the line passes through the mean point, here (2.5, 1.5). If y is already in the column space, $e = 0$, as in the worked example. In NumPy, `np.linalg.solve(X.T @ X, X.T @ y)` gives (2, -0.2).

11 A singular $X^T X$ gives many $\hat{\theta}$ but one $\hat{y}$



$X^T X$ is invertible exactly when the columns of X are independent, rank $M + 1$ (the maths sheet), which needs $N \geq M + 1$. If every x is equal, the x column is a multiple of the ones column. Multicollinearity, basis functions and dummy variables: the linear regression II sheet.

12 Pause and think

What does least squares do geometrically? It projects y onto the column space of X .

Why a column of ones? It carries θ_0 . Without it the line must pass through the origin.

When is $\hat{\theta}$ unique? When $X^T X$ is invertible: independent columns of X .

$\hat{y} = 2 + x$ leaves residuals summing to -8. Least squares? No: with an intercept they sum to 0.

Predicted weight at height 7? $12 + 5.2 \cdot 7 = 48.4$ kg, from the fit in panel 1.

Add a point that lies on the fitted line: does $\hat{\theta}$ change? No: its residual is 0, so $X^T e = 0$ still holds.

$$\text{minimise } \varepsilon^T \varepsilon \cdot X^T X \hat{\theta} = X^T y \cdot \hat{\theta} = (X^T X)^{-1} X^T y \text{ for independent columns} \cdot \hat{y} \text{ projects } y: X^T (y - \hat{y}) = 0$$

Before: the maths for ML sheet; next: the linear regression II sheet. Source: lecture slides "Linear regression" and the linear-regression and geometric-linear-regression notebooks (nipunbatra.github.io/ml-teaching). The height fit, its predictions and the equal- x example are computed here.